

## REVIEW ARTICLE

# ‘To be, or not to be’—The dilemma of ‘silent’ antimicrobial resistance genes in bacteria

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**Abstract**

Antimicrobial resistance is a serious threat to public health that dramatically undermines our ability to treat bacterial infections. Microorganisms exhibit resistance to different drug classes by acquiring resistance determinants through multiple mechanisms including horizontal gene transfer. The presence of drug resistance genotypes is mostly associated with corresponding phenotypic resistance against the particular antibiotic. However, bacterial communities harbouring silent antimicrobial resistance genes—genes whose presence is not associated with a corresponding resistant phenotype do exist. Under suitable conditions, the expression pattern of such genes often revert and regain resistance and could potentially lead to therapeutic failure. We often miss the presence of silent genes, since the current experimental paradigms are focused on resistant strains. Therefore, the knowledge on the prevalence, importance and mechanism of silent antibiotic resistance genes in bacterial pathogens are very limited. Silent genes, therefore, provide an additional level of complexity in the war against drug-resistant bacteria, reminding us that not only phenotypically resistant strains but also susceptible strains should be carefully investigated. In this review, we discuss the presence of silent antimicrobial resistance genes in bacteria, their relevance and their importance in public health.

**KEYWORDS**

antimicrobial resistance, cryptic genes, gene expression, gene silencing, silent genes

## INTRODUCTION

The waxing and waning of the ongoing SARS CoV 2 COVID-19 pandemic have taught us important lessons. Among them, the most relevant is the importance of being informed and being scientifically prepared so that we can mitigate the threat as early as possible. This review is not, however, on Covid-19, but the ‘silent’ second one or the one that may already be upon us—antimicrobial resistance. Antimicrobial resistance (AMR) is defined as the ability of disease-causing microorganisms to resist

the effects of drugs that they were initially susceptible to. The pandemic associated with Covid-19 is devastating. However, the infections associated with AMR bacteria will be completely untreatable, and therefore chaotic.

Globally, AMR in bacterial pathogens represents a serious threat to human, animal and environmental health. AMR imperils the worthy use of antibiotics, and without intervention, humanity is heading for a pre-antibiotic era where even minor infections can be fatal. Even though the indiscriminate use of antibiotics in healthcare settings is apparent, unrestricted use of antibiotics in the food/agriculture

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sector is one of the primary causes determining the AMR evolution in bacteria. In the food sector, antibiotics are either used therapeutically/prophylactically to treat/prevent clinical disease and are used sub-therapeutically to enhance animal growth. Although certain countries have banned the use of antibiotics in food animals (Maron et al., 2013), its semirestricted use in many of the countries might nullified the effect of ban. Overall statistics confirm that food animals are administered with twice more antibiotics than humans. A linear correlation has been established between indiscriminate antibiotic use and the development of AMR in pathogenic micro-organisms (Aslam et al., 2018).

It is estimated that infections associated with AMR cause more than 99,000 human fatalities annually, costing up to \$29,000/patient and associated loss of about \$35 billion annually *w.r.t* (with respect to) the loss of productivity ([https://www.who.int/medicines/areas/rational\\_use/AMR\\_Surveillance/en/](https://www.who.int/medicines/areas/rational_use/AMR_Surveillance/en/)). The recently commissioned first economic report on the impact of AMR concluded that, by 2050, deaths associated with AMR pathogens will increase to 10 million/annum overtaking cancer as the main cause of mortality (<https://amr-review.org>).

Micro-organisms adopt different strategies to tolerate antimicrobials—(1) reduced permeability or drug uptake due to modifications in the porin proteins, (2) increased efflux of the antimicrobials due to the activity of efflux pumps, (3) enzymatic inactivation of the antibiotic, (4) alteration of drug target through spontaneous point mutations, (5) loss of enzymes involved in drug activation and use of alternative metabolic pathway to produce target end product (Blair et al., 2015; Karunasagar et al., 2012). The genetic factors encoding these mechanisms are often carried on mobile genetic elements, increasing the risk of conjugational/HGT-associated spread of antimicrobial resistance among multiple bacterial populations. In most cases, the incoming resistant gene results in causing phenotypic resistance in the recipient bacterial pathogen. However, there have been some sporadic incidences of bacterial isolates harbouring antimicrobial resistance genes while remaining susceptible to the corresponding antimicrobial agent (Deekshit et al., 2012; Enne et al., 2006; Ghanbarpour et al., 2020; Kime et al., 2019; Vk et al., 2019). Such genes, though silent in the carrier strain, if transmitted to a recipient strain can cause resistant phenotype in the latter. In this minireview, we will discuss silent antimicrobial resistance determinants and their biological relevance.

## SILENT ANTIMICROBIAL RESISTANCE GENES

Silent antimicrobial resistance genes (cryptic genes) are defined as those genes that are carried by the bacteria

either on a plasmid or chromosomal DNA without exhibiting corresponding phenotypic resistance to the antibiotics (Kime et al., 2019; Tamburini & Mastromei, 2000). Such cryptic genes are sporadically identified in bacterial pathogens (Hall, 1999; Parker & Hall, 1990). The WHO priority list of pathogens such as *Klebsiella pneumoniae*, pathogenic *E. coli*, *Salmonella* spp. and *Acinetobacter baumannii* is known to harbour silent resistance genes for different classes of antimicrobial agents with varying levels of prevalence (Stasiak et al., 2021). For example, 16% of *A. baumannii* strains harboured a silent *bla*<sub>OXA-23</sub> and remained sensitive to imipenem (Carvalho et al., 2011). Similarly, 28% of *E. coli* carried *aadA* gene (Lanz et al., 2003) while 3% of *Salmonella* spp. harboured *aadA* (Ma et al., 2007). A very low percentage of colistin susceptible *E. coli* isolates also carried *mcr-1* gene (Fernandes et al., 2016).

Similarly, the presence of *bla*<sub>CTX</sub>, *bla*<sub>SHV</sub> and *bla*<sub>TEM</sub> genes in some clinical *Klebsiella pneumoniae* strains did not confer resistance to a cephalosporin (Xu et al., 2014). In the above cases, the genes had intact promoter and resistance gene sequences suggesting that the mechanism of silencing is uncharacterized. For such genes with an intact genetic makeup, silencing was shown to be very conditional—that is, the genes that were silent in one condition were activated in other conditions. A typical example is that of the resistance genes *bla*<sub>OXA-2</sub>, *aadA1*, *sul1* and *tetA* carried on the plasmid pVE46 of a porcine *E. coli* isolate (Enne et al., 2006). In the original strain, the phenotypic expression of resistance conferred by the corresponding gene was intact. However, when the strain was infected orally into pigs, a subset of the isolates recovered from faecal samples did not express resistance, despite possessing an intact plasmid. Transformation of the plasmid from the silenced strain to a new isolate restored gene expression suggesting that the antimicrobial resistance genes were silenced *in vivo*. This suggests that harsh conditions like gut allow genes to be transmitted silently among bacterial populations and, in the future, express the phenotypic resistance under selection pressure. In other cases, the silent genes could become active when transferred into a new host—for example, the gene *aadA*, silent in Shiga toxin-producing *E. coli* was fully expressed only after its transfer into *Hafnia alvei* (Zhao et al., 2001). Even the medium in which the bacterium was grown affected the gene silencing. Silent genes can also be activated by mutations leading phenotype variation or by growing the strain in glucose/glycerol minimal media. A clinical *Salmonella* Typhimurium strain had a cryptic *aadA* gene, encoding aminoglycoside adenyl transferase (Koskiniemi et al., 2011). During growth in a rich medium, the strain was susceptible to streptomycin and spectinomycin but when shifted to glucose/glycerol

minimal media, the *aadA* gene was activated resulting in streptomycin/spectinomycin resistance. Here, the amino acid or carbon starvation induced the bacteria to produce the alarmone guanosine penta/tetraphosphate (ppGpp) which acted as a positive regulator for *aadA*. Interestingly, growth in the glucose/glycerol minimal media resulted in the production of small colony variants which had stop codon variants in the genes *hemA* and *ubiA*. For these small colony variants, streptomycin stimulated growth by inducing read-through in these stop codons (Koskineniemi et al., 2011).

In some cases, the genetic makeup of the cryptic genes was not intact, leading to gene silencing. A small proportion of seafood isolated *Salmonella* Weltevreden strains carried chloramphenicol acetyltransferase (*catA1*) gene did not express the resistance (Deekshit et al., 2012). In this case, the absence of a promoter region was found to be the main reason for the phenotypic susceptibility of the isolates. The presence of a plasmid-mediated quinolone resistance (PMQR) *qnrB* gene was noted in clinically isolated *E. coli* strains that were sensitive to quinolone/fluoroquinolones (6), suggesting the silent nature of the gene. This is, however, debatable as the presence of PMQR genes usually encodes only a low-level resistance to fluoroquinolones while quinolone/fluoroquinolone resistance is mainly attributed to the occurrence of spontaneous mutations at the quinolone resistance determining regions (Shetty et al., 2019).

Most of the strains harbouring silent genes are clinical, pointing out that hospitals can act as reservoirs of silent genes and these genes can spread between bacterial populations (Zhao et al., 2001). Besides, many antimicrobial resistance determinants present in natural environments are cryptic (Canton, 2009). The silent genes can be transferred via HGT, like normal genes do (Hall et al., 1983). Overall, the presence of silent genes could be a very common phenomenon but could play a very important role in bacterial adaptation to the environment and their evolution (Hall et al., 1983). Silent antimicrobial resistance genes, therefore, are an additional layer of threat in antimicrobial therapy. The silent resistance genes and their mechanisms are summarized in Table 1.

## WHY ARE WE MISSING OUT ON SILENT GENES?

The assays which are currently implemented in antimicrobial studies are very biased towards resistant strains. For example, assays involving minimum inhibitory concentrations (MIC), disk diffusion assays recognize only resistant isolates. Even molecular analyses like

PCR, PFGE, and WGS are mostly used to characterize the molecular basis of resistance in drug-resistant strains. Sensitive strains, harbouring silent genes, are mostly ignored. Furthermore, the phenotype of the silent resistance gene may remain masked in the presence of another active resistance gene in the genome. For example, tetracycline or chloramphenicol resistance can be mediated by single or multiple *tet* or *cat* genes respectively. The resistant strain can harbour either a single gene or multiple genes encoding the resistance. In majority of such cases, the isolate harbouring multiple resistance genes encodes higher level of resistance than those with single gene. This is demonstrated in a recent study (Jahantigh et al., 2020) where *E. coli* strains carrying *tetD* remained sensitive while either *tetA* and/or *tetB* genes significantly correlated with resistance. This reiterates the importance of decoding the role of individual resistance genes in conferring resistance towards a particular antibiotic irrespective of the number of genes carried by the isolates. Therefore, to identify and characterize silent AMR genes and to avoid a potential therapeutic failure, we need to include both resistant and sensitive strains in the paradigm of AMR studies, particularly those assays using PCR and deep genome sequencing technologies.

## WHY DO BACTERIA SILENCE ANTIMICROBIAL RESISTANCE GENES?

Bacteria always tend to silence genes that are xenogeneic (Navarre, 2016). It distinguishes xenogeneic and ancestral genes. A bacterium uses base composition differences to distinguish foreign and self-genes since xenogeneic genes tend to be more AT-rich than the rest of the genome. The typical example is the pathogenicity islands of *Salmonella* (Desai et al., 2013). From the bacterial perspective, silencing is very essential as the organism is unaware of the fitness cost exerted on the host cell if the xenogeneic sequence is expressed without regulation. This is demonstrated in *Salmonella*, as an uncontrolled expression of the *Salmonella* Pathogenicity Island 1 was responsible for the reduced growth rate of a *Salmonella* *hns* mutant (Ali et al., 2014). To silence xenogeneic genes, bacteria rely on classic silencing proteins such as H-NS, Lsr2 and MvaT, and additional factors like Hha/YdgT, who selectively recognize and bind to the minor groove of the xenogeneic AT-rich DNA (Madrid et al., 2007; Navarre, 2016; Solorzano et al., 2015).

Antimicrobial resistance genes are mostly xenogeneic and obtained via horizontal gene transfer (HGT). The presence of antimicrobial resistance genes is known to

TABLE 1 Reported silence resistance determinants in bacteria

| Class of antibiotic            | Antibiotic tested                | Resistance gene            | Bacterial strains                              | Mechanism                                                                    | References             |
|--------------------------------|----------------------------------|----------------------------|------------------------------------------------|------------------------------------------------------------------------------|------------------------|
| Phenicol                       | Chloramphenicol                  | <i>catA1</i>               | <i>Salmonella</i> Weltevreden                  | Promoter deletion                                                            | Deekshit et al. (2012) |
|                                |                                  | <i>catA2</i> and <i>A3</i> | <i>S. Wilhelmsburg</i> , <i>S. Enteritidis</i> | Unknown                                                                      | Matchawe et al. (2021) |
| Quinolones/<br>fluoroquinolone | Nalidixic acid and ciprofloxacin | <i>cmfA 1, 5 and 6</i>     | <i>S. Enteritidis</i>                          |                                                                              |                        |
|                                |                                  | <i>qnrB</i>                | <i>Escherichia coli</i>                        | Unknown (possible presence of prophage)                                      | Cayo et al. (2017)     |
| Aminoglycosides                | Gentamycin                       | <i>aacA-aphD</i>           | <i>Staphylococcus aureus</i>                   | Disruption of coding region by the insertion of IS256                        | Kime et al. (2019)     |
|                                | Tobramycin                       | <i>aacA-aphD</i>           | <i>S. aureus</i>                               | Disruption of coding region by the insertion of IS256                        | Kime et al. (2019)     |
| Spectinomycin                  |                                  | <i>ant4</i>                | <i>S. aureus</i>                               | Frameshift mutation due to the insertion of single nucleotide                | Kime et al. (2019)     |
|                                |                                  | <i>aac (6')-I</i>          | <i>S. Enteritidis</i>                          | Presence of <i>sgc</i> cluster                                               | Magnet et al. (1999)   |
|                                |                                  | <i>aad9</i>                | <i>S. aureus</i>                               | Frameshift mutation due to the deletion of single nucleotide in poly A tract | Kime et al. (2019)     |
|                                |                                  | <i>aad9</i>                | <i>S. aureus</i>                               | Nonsense mutation due to nucleotide substitution                             | Kime et al. (2019)     |
|                                |                                  | <i>aadA1</i>               | <i>S. Typhimurium</i>                          | Unknown                                                                      | Ma et al. (2007)       |
|                                |                                  | <i>aadA2</i>               | <i>S. Choleraesuis</i>                         | Unknown                                                                      | Ma et al. (2007)       |
|                                |                                  | <i>strAB</i>               | <i>E. coli</i>                                 | Part of <i>strB</i> gene deleted                                             | Enne et al. (2006)     |
|                                |                                  | <i>aadA1</i>               | <i>E. coli</i>                                 | Unknown                                                                      | Enne et al. (2006)     |
|                                |                                  | <i>aadA2</i>               | <i>E. coli</i>                                 | Unknown                                                                      | Enne et al. (2006)     |
|                                |                                  | <i>ant (6)-Ia</i>          | <i>B. anthracis</i> BR4253                     | Probable repressor activity                                                  | Perreten et al. (2005) |
| Streptomycin                   |                                  | <i>aadA</i>                | <i>S. Wilhelmsburg</i>                         | Unknown                                                                      | Matchawe et al. (2021) |
|                                |                                  | <i>aadA1</i>               | <i>S. Enteritidis</i>                          |                                                                              |                        |
|                                |                                  | <i>aadA2</i>               | <i>S. Wilhelmsburg</i>                         |                                                                              |                        |
|                                |                                  |                            | <i>S. Infantis</i>                             |                                                                              |                        |
|                                |                                  |                            | <i>E. coli</i>                                 |                                                                              |                        |
| Kanamycin and neomycin         |                                  | <i>Aph(3')-Ia</i>          | <i>Bacillus anthracis</i> BR4253               | Unknown                                                                      | Maynard et al. (2003)  |
|                                |                                  | <i>aph(3')-III</i>         |                                                | Probable repressor activity                                                  | Perreten et al. (2005) |

(Continues)

TABLE 1 (Continued)

| Class of antibiotic | Antibiotic tested             | Resistance gene                                                                | Bacterial strains              | Mechanism                                                                                     | References              |
|---------------------|-------------------------------|--------------------------------------------------------------------------------|--------------------------------|-----------------------------------------------------------------------------------------------|-------------------------|
| $\beta$ -lactams    | cefotaxime                    | <i>mecA</i>                                                                    | <i>S. aureus</i>               | Frameshift mutation due to the insertion of single nucleotide                                 | Kime et al. (2019)      |
|                     |                               | <i>mecA</i>                                                                    | <i>S. aureus</i>               | Truncated <i>bla<sub>RI</sub></i> regulator gene causing repression of <i>mecA</i> expression | Kime et al. (2019)      |
|                     | Penicillin                    | <i>bla<sub>Z</sub></i>                                                         | <i>S. aureus</i>               | Frameshift mutation due to the deletion of single nucleotide in play A tract                  | Kime et al. (2019)      |
|                     | Imipenem                      | <i>cfiA</i>                                                                    | <i>Bacteroides fragilis</i>    | Unknown                                                                                       | Podglajen et al. (1994) |
|                     |                               | <i>bla<sub>IMP</sub></i>                                                       | <i>Klebsiella pneumoniae</i>   | Unknown                                                                                       | Walsh (2005)            |
|                     | Ertapenem                     | <i>bla<sub>CTX</sub></i> , <i>bla<sub>SHV</sub></i> , <i>bla<sub>TEM</sub></i> | <i>K. pneumoniae</i>           | Unknown                                                                                       | Xu et al. (2014)        |
|                     | Amikacin                      | <i>bla<sub>OXA-23</sub></i>                                                    | <i>Acinetobacter baumannii</i> | Unknown                                                                                       | Carvalho et al. (2011)  |
|                     | Penicillin                    | <i>bla<sub>1</sub></i> and <i>bla<sub>2</sub></i>                              | <i>B. anthracis</i>            | The repressor, <i>BlaI</i> , blocks transcription of the $\beta$ -lactamase gene              | Chen et al. (2003)      |
|                     | Ampicillin                    | <i>bla<sub>TEM-1</sub></i> , <i>bla<sub>TEM-163</sub></i>                      | <i>S. Enteritidis</i>          | Unknown                                                                                       | Matchawe et al. (2021)  |
|                     |                               | <i>bla<sub>TEM-163</sub></i>                                                   | <i>S. Infantis</i>             |                                                                                               |                         |
|                     | Meropenem                     | <i>bla<sub>IMP-16</sub></i>                                                    | <i>Pseudomonas aeruginosa</i>  | GGG deletions in the promoter region of the integron                                          | Picao et al. (2012)     |
|                     |                               |                                                                                |                                |                                                                                               |                         |
| Lincosamides        | Clindamycin                   | <i>vga(A)<sub>v</sub></i>                                                      | <i>S. aureus</i>               | Possibly due to low copy number of Tn5406                                                     | Kime et al. (2019)      |
|                     |                               | <i>ermA</i>                                                                    | <i>S. aureus</i>               | Loss of translation initiation sequence                                                       | Kime et al. (2019)      |
|                     | Erythromycin                  | <i>ermA</i>                                                                    | <i>S. aureus</i>               | Disruption of coding sequence due to the insertion of IS256                                   | Kime et al. (2019)      |
|                     |                               | <i>ermA</i>                                                                    | <i>S. aureus</i>               | Nonsense mutation due to nucleotide substitution                                              | Kime et al. (2019)      |
| Streptogramins      |                               | <i>ermA</i>                                                                    | <i>S. aureus</i>               | Missense mutation due to nucleotide substitution                                              | Kime et al. (2019)      |
|                     | Quinupristin and dalbapristin | <i>vga(A)<sub>v</sub>/ermC</i> , <i>vga(A)<sub>v</sub>/ermA</i>                | <i>S. aureus</i>               | Unknown                                                                                       | Kime et al. (2019)      |

TABLE 1 (Continued)

| Class of antibiotic | Antibiotic tested | Resistance gene         | Bacterial strains                                                            | Mechanism                                                               | References              |
|---------------------|-------------------|-------------------------|------------------------------------------------------------------------------|-------------------------------------------------------------------------|-------------------------|
| Tetracyclines       | Tetracycline      | <i>tetK</i>             | <i>S. aureus</i>                                                             | Separation of promoter from coding region due to the insertion of IS431 | Kime et al. (2019)      |
|                     |                   | <i>tetM</i>             | <i>S. aureus</i>                                                             | Frameshift mutation due to the deletion of single nucleotide            | Kime et al. (2019)      |
|                     |                   | <i>tetB</i>             | <i>Streptococcus pyogenes</i>                                                | Unknown                                                                 | Brenciani et al. (2007) |
|                     |                   | <i>tetB, tetC, tetG</i> | <i>E. coli</i>                                                               | Part of the gene is missing                                             | Enne et al. (2006)      |
|                     |                   | <i>tetR</i>             | <i>S. Wilhelmsburg, S. Infantis, S. Wernigerode</i>                          | Unknown                                                                 | Matchawe et al. (2021)  |
| Sulphonamides       | Sulphonamide      |                         | <i>S. Enteritidis, S. Wilhelmsburg, S. Infantis S. Wernigerode S. Kibusi</i> | Unknown                                                                 | Matchawe et al. (2021)  |
|                     |                   | <i>Sul2</i>             | <i>E. coli</i>                                                               | Part of the gene is missing                                             | Enne et al. (2006)      |
| Polymyxin           | Colistin          | <i>mcr-1</i>            | <i>Vibrio parahaemolyticus</i>                                               | Unknown                                                                 | Jiang et al. (2014)     |
| Trimethoprim        | Trimethoprim      | <i>dfrA17</i>           | <i>E. coli</i>                                                               | Insertion of IS1294b                                                    | Zhou et al. (2018)      |
| Miscellaneous       | Mupirocin         | <i>mupA</i>             | <i>E. coli</i>                                                               | Absence of promoter                                                     | Enne et al. (2006)      |
|                     |                   |                         | <i>S. aureus</i>                                                             | Frameshift mutation due to the deletion of single nucleotide            | Kime et al. (2019)      |



reduce bacterial fitness (Andersson et al., 2007; Andersson & Hughes, 2010), even though compensatory mutations elsewhere in the genome can eliminate this fitness cost. In the former case, lesser fit antimicrobial-resistant bacteria can be outcompeted by more fit antimicrobial sensitive bacteria during growth in an antimicrobial-free medium. Theoretically, there are three ways a bacterial host can counteract the potential fitness cost exerted by the presence of antimicrobial resistance genes, especially when they are on mobile genetic elements—(1) acquire compensatory mutations, (2) loss of the mobile genetic element, (3) silencing the gene by switching off its expression. The silencing of plasmid-borne antimicrobial resistance genes had a beneficial effect on the fitness of the host even though this effect was dependent on the nature of the plasmid (Humphrey et al., 2012).

The most well-characterized mechanism by which bacteria 'silence' its genes, especially those obtained xenogenetically via HGT, is by employing silencing proteins. The classic and most well-studied silencing protein is the Heat Stable Nucleoid Structuring Protein (H-NS) and H-NS-like proteins like Lsr2 and MvaT (Navarre, 2016). These proteins insert themselves into the minor groove of AT-rich DNA using the 'AT-hook' (H-NS/Lsr2) or 'AT-pincer' (MvaT) and block RNA polymerase from initiating transcription from the xenogeneic sequences. Once xenogeneic sequences are acquired and incorporated, bacteria employ these silencing proteins to temporarily silence these genes by switching off their expression. However, as well characterized in the case of pathogenicity islands of *Salmonella*, there are counter silencers/transcriptional activators that switch 'on' these H-NS silenced xenogeneic genes at the 'right time/right place' so that the host bacterium benefits from their expression. So, one plausible explanation is that these antimicrobial resistance genes are under temporary silencing by these silencing proteins and could be activated by counter silencing transcriptional factors at a time when the phenotype is most beneficial.

## WHY ARE CRYPTIC GENES MAINTAINED IN THE BACTERIAL GENOME?

It was proposed very early that mutational activation of cryptic genes forms a basis for microbial evolution and adaptation (Hall et al., 1983). This study also used mathematical modelling to predict that biologically important mechanisms exist to preserve the cryptic genes on the genome. Doubts were later cast on the biological importance of cryptic genes when it was shown that cryptic genes are silent only in laboratory conditions and their expression is induced in the natural environment (Tamburini &

Mastromei, 2000). The authors also conclude that these genes are rare and encode unusual functions. However, this was quickly disproven when it was shown that silent forms of genes were shown to be not so rare and could be a source of new resistant phenotypes (Lanz et al., 2003). Besides, silent genes can spread among different bacterial genera and can become active in other bacterial genera (Zhao et al., 2001). In the natural environment, AMR genes enable bacteria to survive natural antimicrobials present in the environment. This AMR gene pool, otherwise called 'resistome', can be mobilized via mobile genetic components from bacteria in the natural environments to pathogenic bacteria, who can incorporate the genes and express them to facilitate resistance when needed. Silent genes could be widespread in the resistome, ready for recall by mutational activation in the future bacterial generations (Hall et al., 1983). The cryptic genes could function as a genetic reservoir that enhances the adaptive potential of the bacterial species by a mechanism that is independent of genetic exchange. Silent genes, therefore, provide an additional level of complexity, reminding us that not only phenotypically resistant strains but also susceptible strains should be carefully investigated. If the potential of silent genes is ignored, it will lead to underestimating the overall potential of antimicrobial resistance in the bacterial population, leading to a potential therapeutic failure.

## MECHANISM OF ANTIBIOTIC RESISTANCE GENE SILENCING

### Silencing of antibiotic resistance by mutation (SARM)

SARM is one of the most important mechanisms responsible for the silencing of bacterial genes (Kime et al., 2019). Of 1470 *Staphylococcus aureus* isolates tested, ~10% harboured silent resistance genes while 3.1% exhibited gene silencing via the SARM mechanism. Here, the presence of mutations within/upstream antibiotic resistance determinants silenced the expression of the resistance gene. Quite understandably, mutations affected the genes by multiple mechanisms—disruption of the coding sequence, causing frameshift, nonsense mutation, loss of translation initiation elements, a missense mutation in the binding pocket for co-substrate, separation of the promoter from coding sequence and truncation of regulator gene causing the repression antibiotic resistance gene (Kime et al., 2019; Wong et al., 2009). Similarly, some of the isolates harbouring fully intact functional *vga(A)*<sub>v</sub> gene did not show resistance to clindamycin. This may be due to insufficient expression of *vga(A)*<sub>v</sub>, due to the low copy number of transposable elements Tn5406, rather than the SARM mechanism (Tesse et al., 2013).

SARM can also be either reversible or irreversible. In reversible mechanism, the phenotypic resistance may be regained and irreversible mechanism involves permanent loss of antibiotic resistance. A reversible SARM mechanism was demonstrated in a majority of *S. aureus* isolates as they regained antibiotic resistance at a frequency of  $>10^9$  cells during infection (Kime et al., 2019). In general, SARM includes a group of hidden antibiotic resistance genes that evades routine normal detection process and has similarities with a phenomenon called heteroresistance (Nicoloff et al., 2019). Even though there is no single point solution to address this issue, deep sequencing of bacterial genomes and metagenomics could help in the identification and characterization of silent AMR genes.

## Promoter region effect

Another important factor that results in the silencing of the resistance gene would be the alterations/deletions within the promoter region. For instance, the presence of the cryptic *catA1* gene in *S. Weltevreden* carried on a plasmid can be related to large deletion upstream of the antibiotic resistance gene (Deekshit et al., 2012). Similarly, the silencing of *aac(6')-I* gene can also be attributed to the deletion in the promoter region (McMillan et al., 2019). Among the two clinical *Salmonella* Enteritidis strains (BM4361 and BM4362) isolated from the same patient, one harboured a cryptic aminoglycoside resistance gene *aac(6')-Iy* with tobramycin susceptibility and the second strain with resistance to tobramycin (Magnet et al., 1999). The tobramycin resistance was due to a 60-kb deletion upstream to the *aac(6')-Iy* gene which caused transcriptional fusion leading to the phenotypic expression of resistance.

## Presence of integrons

Integrons are mobile genetic elements capable of integrating a large number of resistance gene cassettes with the help of integrase (Carattoli, 2001). They are the natural expression systems that can integrate ORFs and convert them into functional genes. Structurally they consist of basic units including an integrase gene (*intI*), attachment site (*attI*) and a strong promoter to facilitate the expression of integrated genes. The site *attI* is capable of integrating a large number of promoter-less antibiotic resistance gene cassettes and can express all the integrated genes with a single strong promoter. Five groups of integrons have been described so far, namely, (1) *Class 1 integron* is one of the most widely spread mobile integrons among Gram-negative bacterial pathogens. It consists of a 5' conserved segment (CS) attached with

integrase gene (*intI*), *attI* site with a strong promoter while 3' CS consists of *sulI* coding for sulphonamide resistance, a quaternary ammonium compound resistance *qacEΔ1* gene, and ORF5 of unknown function (Deekshit et al., 2012; Recchia & Hall, 1995). (2) *Class 2 integrons* are exclusively associated with *Tn7* transposons with three gene cassettes (*dhfr1-sat-aadA1*) coding for aminoglycoside resistance adjacent to an integrase gene *intI2* within 5' CS (Radstrom et al., 1994). (3) *Class 3 integron* is less prevalent carrying *bla<sub>IMP</sub>* conferring resistance to β-lactamases and a part of *aacA4* gene and an *intI3* gene associated with a transposon was identified in some bacteria (Arakawa et al., 1995; Collis et al., 2002). (4) *Class 4 integron* with *intI4* integrase gene has been identified in *Vibrio cholerae* in association with an integron island on a small chromosome consisting of repeated sequences (Hochhut et al., 2001). (5) *Class 5 integron* identified on a transposon in pRSV1 plasmid of *Aliivibrio salmonicida* (Sorum et al., 1992).

Insertion of IS elements can lead to the inactivation/activation of drug resistance genes. The alarming case is that of an *mcr-1* gene that encodes resistance against colistin. The *mcr-1* was found to be inactivated in a fluoroquinolone-resistant and colistin-sensitive *E. coli* strain due to the insertion of a 1.7-kb insertion sequence *IS1294b* within the gene. After multiple passages, the disrupted gene was reactivated due to the loss of *IS1294b*—an effect that has potential for wider dissemination with the help of pHNSHP45 plasmids (Zhou et al., 2018). Needless to emphasize here that Colistin, being one of the drugs of last resort, the 'silent yet resuscitable' *mcr-1* gene carried on a transferable plasmid is a direct threat to successful therapy. However, upstream insertion of IS elements resulted in the direct activation of silent antimicrobial resistance genes (Depardieu et al., 2007). A typical example is that of the carbapenem resistance gene *cfiA* which was detected in a small number of the human intestinal anaerobic pathogen *Bacteroides fragilis* (Podglajen et al., 1994). The transcription of *cifA*, generally silent, was activated by the presence of two insertion sequences (*IS1187* and *IS1188*) within a 90 base pair region upstream *cfiA* (Podglajen et al., 2001). Such transposition of certain IS elements created a strong hybrid promoter that activated the expression in silent genes. Similarly, *IS18* transposition in *Acinetobacter* sp. 13 strain BM2716 created a hybrid promoter that activated the expression of the otherwise silent aminoglycoside resistance gene *aac(6')-Ij* (Rudant et al., 1998)—resulting in the strain becoming aminoglycoside resistant. The transposition of *IS257* induced high- and low-level trimethoprim (*dhfrA*) and tetracycline (*tetA [K]*) resistance in *S. aureus* (Leelaporn et al., 1994; Simpson et al., 2000). Insertional inactivation of *oprD* by *ISPa1328*, *ISPa1635* in clinical



isolates of *Pseudomonas aeruginosa* resulted in increased carbapenem resistance (Wolter et al., 2004).

All the gene cassettes integrated into the integron structure are constitutively expressed with the help of a single strong promoter (Pc) (Jove et al., 2010). However, the distance of the gene from the promoter can also determine gene expression, as exemplified in *A. baumannii* (Coyne et al., 2010) where three resistance gene cassettes are identified in the class 1 integron on AbaR1 island but showed reduced expression. Therefore, even though integrons are capable of acquiring a large number of resistance genes (sometimes up to 60 genes), many of these genes are promoterless and might remain unexpressed even in the presence of the strong integron-specific promoter Pc (Cambray et al., 2010). Besides, free gene cassettes lacking an integron are usually silent because the integron's promoter is required for their expression (Carattoli, 2001; Enne et al., 2006).

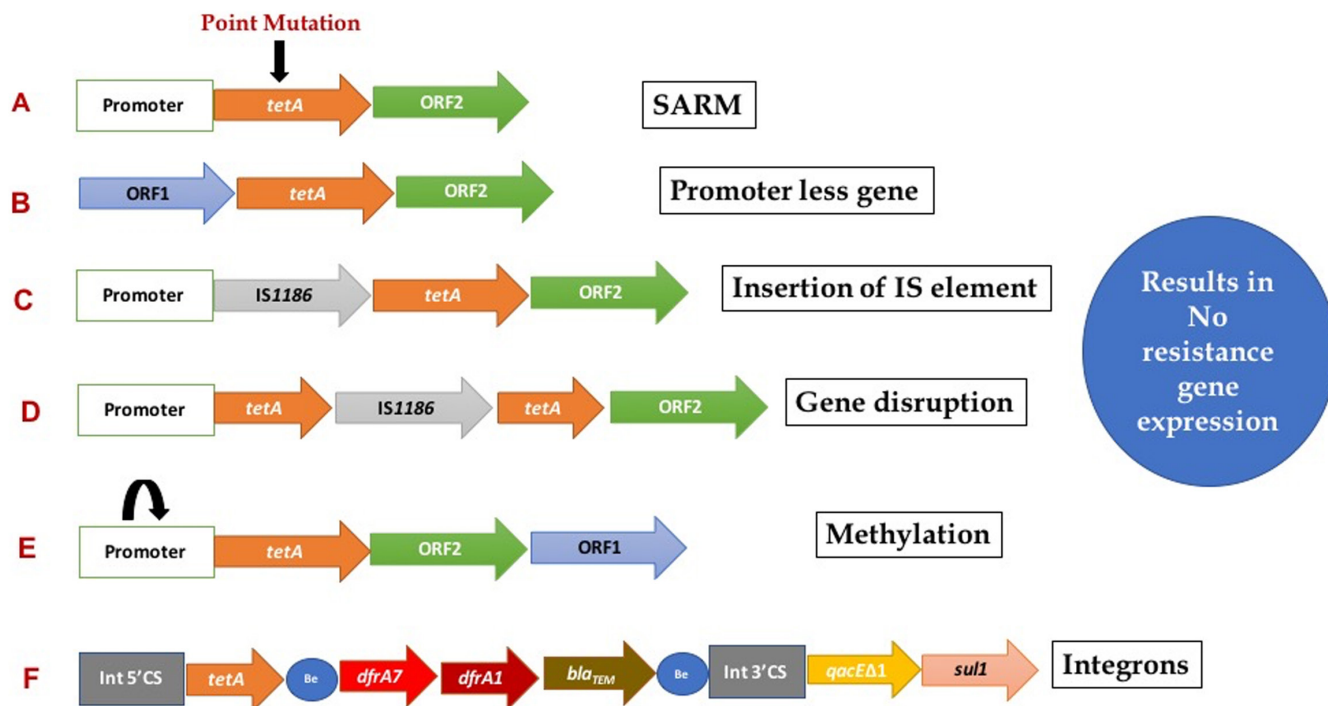
## Regulatory factors

The role of regulatory factors in gene silencing has been lately explored in bacterial pathogens (Navarre, 2016). Many genes may remain silent due to strong negative transcriptional regulators. For instance, oxacillin susceptible *S. aureus* carrying a *mecA* gene has been reported recently from many countries including Germany, the United Kingdom, Greece, India, Taiwan and Japan (Pournaras et al., 2013). However, in MRSA (methicillin-resistant *S. aureus*), the *mecA* gene expression is usually modulated by  $\alpha$  DNA-binding protein (MecI) and an  $\alpha$  signal transducer (MecR1) respectively. The corresponding genes are organized in an operon upstream of *mecA* in the SCC<sub>mec</sub> element (Hiramatsu et al., 1992). The repressor MecI represses the transcription of *mecA* and *mecI-mecR1* in the absence of  $\beta$ -lactam antibiotics by binding to two palindromes located within the operon (Safo et al., 2006). In the presence of  $\beta$ -lactam antibiotics, MecR1 binds with the antibiotic and de-repression occurs. Consequently, the altered structure of MecR1 results in activation of metalloprotease and causes cleavage of MecI, inactivating the repressor leading to the synthesis of penicillin-binding protein 2 $\alpha$  (PBP2 $\alpha$ ) (Marrero et al., 2006). Besides, the expression of *mecA* can also be regulated by the *bla* system which is homologous to the *mec* system. The *blaI-blaR1* system regulates the synthesis of  $\beta$ -lactamase (BlaZ) and thereby confers resistance to penicillin (Clarke & Dyke, 2001). The set of genes *blaI-blaR1-blaZ* is either carried on a plasmid or a transposon. The similarity between the *bla* and *mec* system allows substantial cross-talk between them and facilitates the binding of *blaI* to the *mecA* operator region to prevent the transcription of *mecA*.

Genome sequencing revealed the presence of a functional *BlaI* repressor and a nonfunctional *BlaR1* transducer carried on a plasmid to be responsible for oxacillin susceptibility in *mecA*-positive *S. aureus* (Sabat et al., 2015). However, such silent genes can always recover their ability to show phenotypic resistance by two distinct mechanisms; namely, Excision of transposable element IS1181 from *mecA* and slip strand replication errors within *mecA* (Proulx et al., 2016). Nevertheless, excision of transposable element works by induction to give rise to revertant, while slipped strand replication errors are of more concern because it arises from sequence properties of *mecA* genes. In bacteria, the continuous cycle of phase variation of positive and negative selection acting on expression of a particular gene results in genetic variations that leads to the production of individuals with silent resistance genes that in turn mediated my slipped strand mutagenesis (van der Woude & Baumler, 2004). This phenomenon is observed in *S. aureus* strains that transformed from *mecA* -positive MSSA to MRSA in vivo (Proulx et al., 2016). Since very limited data are available on the biology of such silent resistant *Staphylococcus* isolates and their ability to revert into fully functional forms by regaining all regulatory factors in the clinical set-up, it is important to combine the culture-based sensitivity assays with the molecular detection methods to conclude the true nature of the isolates.

## Epigenetic changes

Epigenetic changes though are not directly involved in silencing adjacent genes by deletion or mutation, they alter the expression pattern of the gene without altering the underlying DNA sequence. DNA methylation is one of the important epigenetic changes that have a profound effect on gene expression including antibiotic resistance genes (Ghosh et al., 2020). By definition, DNA methylation is the addition of a methyl (CH<sub>3</sub>) group to the cytosine residue of the DNA that evolved as a host defence mechanism to degrade foreign DNA such as transposable elements, viral DNA and other exogenous DNA sequences (Bird, 1986). The excessive DNA methylation of the promoter region prevents the expression of genes by inhibiting the binding of transcription factors to the promoter. Conversely, increased methylation can repress and further derepress the expression of the resistance gene thereby promoting its expression (Ghosh et al., 2020). The role of DAM methylase gene has also been explored, wherein high expression of DAM methylase was found to increase the survival rate of bacterial cell treated with nalidixic acid by fivefold with twofold increase the level of efflux pump expression (Adam et al., 2008). Therefore, it is hypothesized that efflux pump regulatory network can be a target for epigenetic



**FIGURE 1** Different mechanisms of reduced gene expression.

modifications thereby involved in the conversion of bacterial cell from susceptible to resistance or vice versa (Motta et al., 2015). Different mechanisms of reduced gene expression/gene silencing are represented in Figure 1.

## CONCLUSION

Bacteria are known to adapt to harsh environments (Anvarian et al., 2016; Fierer, 2017; Lehner et al., 2018). Many of the harsh food processing environments promote HGT in bacteria making food/food processing environments a major source of antimicrobial resistance factors for food-borne pathogenic bacteria (Verraes et al., 2013). Once these genes are acquired and incorporated, bacteria can use multiple methods to temporarily silence these genes until they bring the genes under very strict regulatory control. Even though such genes are silent in vitro might get activated during infection, thereby making therapy difficult. The high prevalence of isolates showing SARM mechanism with a high frequency of occurrence of revertant also challenges the present in vitro susceptibility testing as a reliable predictor of bacterial response to an antimicrobial drug in vivo. Thus, detecting phenotypic drug resistance by disk diffusion assay would not be sufficient since they cannot detect silent antibiotic resistance genes that might get expressed in vivo or transmitted to other bacteria if carried on a transferable plasmid (Frye et al., 2006).

Therefore, we emphasize the importance of identifying silent antimicrobial resistance genes and recognizing their wide potential in disseminating antimicrobial resistance in bacteria.

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## CONFLICT OF INTEREST

There is no conflict of interest declared.

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